

Ujjwal Keshri

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SUMMARY

An ambitious electrical engineer with hands-on experience in embedded systems, IoT sensor integration, and power system modelling. Skilled in C/C++ microcontroller programming and MATLAB/Simulink for simulation, real-time control, and data-driven analysis. Passionate about building reliable and intelligent engineering solutions across hardware, software, and cyber-physical domains.

WORK EXPERIENCE

Industrial Trainee

June 2024 - July 2024

Indian Oil Corporation Limited — SERPL, Bhubaneswar Pump Station (Jatni)

- Assisted in the maintenance and troubleshooting of electrical systems and equipment at the SERPL, Bhubaneswar Pump Station.
- Gained practical experience with the operation and control of automated pumping systems within the pipelines division.

PROJECTS

Power Grid Cyber-Attack Detection via V-I Correlation

[Link to Project](#)

- Modelled a 5-bus AC transmission network in MATLAB/Simulink with normalised per-unit baselines to calibrate detection thresholds.
- Implemented `detect_fdi()` classifying system state as Normal / Cyber-Attack / Physical Fault at 50 μ s sample rate using six FDI attack modes and a weighted event-scoring counter.
- Built `classify_physical_fault()` sub-classifier with four fault codes using RMS buffers and phase-unbalance metrics; detects faults within 150 μ s.

Automated Solar Tracking System

[Link to Project](#)

- Built a dual-axis solar tracker on Arduino using four LDRs to compute horizontal and vertical error signals.
- Implemented a TOLERANCE dead-zone and closed-loop PWM control of two servo motors to actively track the light source and stream JSON data over Serial.

IoT Weather Monitoring Station

[Link to Project](#)

- Built an ESP32-based weather station with a BME280 sensor over I²C, publishing JSON readings to an MQTT broker via WiFiManager.
- Implemented a dual-data pathway: Python backend logs to SQLite and broadcasts via WebSockets; dashboard also supports direct ESP32 connection via Web Serial API.

EDUCATION

2023 - present	BTech-Electrical Engineering at Odisha University Of Technology and Research, Bhubaneswar	(CGPA: 8.46/10)
2022	Class 12th CBSE	(88.60%)
2021	Class 10th CBSE	(89.80%)

SKILLS

Technical Skills

Programming Languages	C/C++, Python, JavaScript [React.js], HTML/CSS
Design & Simulation	Altium Designer, AutoCAD Electrical, MATLAB & Simulink
Hardware & Embedded	Arduino, ESP32, STM32, AVR, PCB Design
Data & Dev Tools	SQLite, JSON, Git, ChatGPT, Gemini, Claude

Soft Skills

Problem-Solving, Analytical Thinking, Detail-Oriented, Self-Motivated